EDAVS: Emotion-Driven Audiovisual Synthesis Experience

Sihang Chen*
sihangchen97@outlook.com
Victoria University of Wellington
Wellington, New Zealand

Junliang Chen*
juneleungchan@163.com
Beijing Film Academy
Beijing, China

Xiaojuan Gu†
guxiaojuan@bfa.edu.cn
Beijing Film Academy
Beijing, China

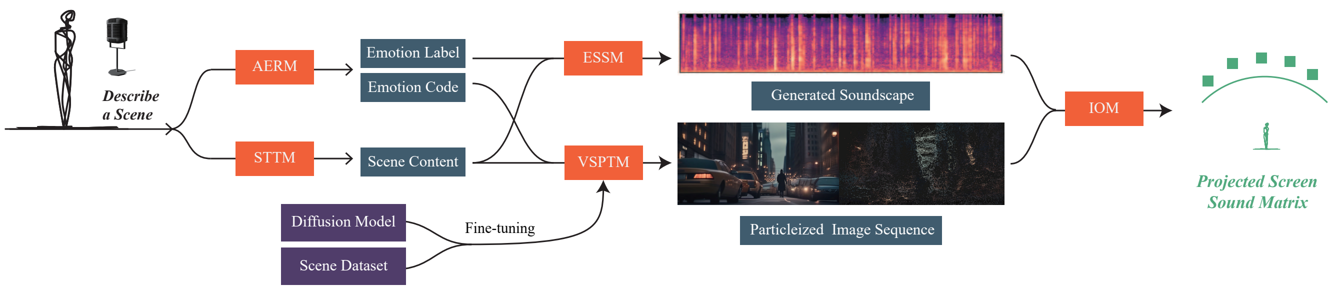

Figure 1: The EDAVS System

ABSTRACT

In this paper, we proposed a novel approach to the creation of immersive multimedia content. At its core, EDAVS harnesses the subtle inflexions of human speech, translating emotive cues into dynamic visual narratives and corresponding soundscapes. This amalgamation of auditory sentiment and visual splendour is predicated upon cutting-edge machine-learning algorithms capable of deep semantic and emotional analysis.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design process and methods**; **Visualization**; • **Computing methodologies** → **Artificial intelligence**.

KEYWORDS

Emotion visualization, Audio-to-Image, Artificial Intelligence Generated Content

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*Equal contributions.
†Corresponding author.

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1 INTRODUCTION

In the ever-changing field of human-computer interaction, affective computing aims to equip machines with the capability to understand and process human emotions. This paper introduces Emotion-Driven Audiovisual Synthesis Experience System (EDAVS), which interprets emotional cues in vocals to generate dynamic visual and auditory displays, facilitating communication between humans and machines.

At the core of EDAVS lies sophisticated emotion recognition and translation from input audio to audiovisual narratives. Drawing inspiration from Picard's foundational work [Picard 2000] and advancements in neural network architectures for real-time emotion analysis [Koolagudi and Rao 2012], our system deciphers not only the semantic content of speech but also its emotion nuances. Building upon multi-modal sentiment analysis [Poria et al. 2017], EDAVS demonstrates proficiency in capturing a range of emotions and rendering them into visual elements, balancing between detailed and abstract forms. This process is complemented by generating a soundscape audio track that harmonizes with the visual content, producing sensory outputs that reflect the user's emotional state.

The significance of EDAVS is multifaceted. In a world where digital content consumption is prevalent, its ability to create emotionally resonant media offers a new perspective on storytelling and digital interaction.

2 SYSTEM DESIGN AND IMPLEMENTATION

The EDAVS system architecture (Fig. 1) performs semantic and emotional analysis of the user's input audio, extracting emotions into a digital spectrum, and rendering visual and auditory content reflecting emotional states. It includes: Audio Emotion Recognition Module (AERM), Speech-to-Text Translation Module (STTM), Visual Synthesis and Particle Transformation Module (VSPTM), Emotive Soundscape Synthesis Module (ESSM), and Integration and Output Module (IOM).

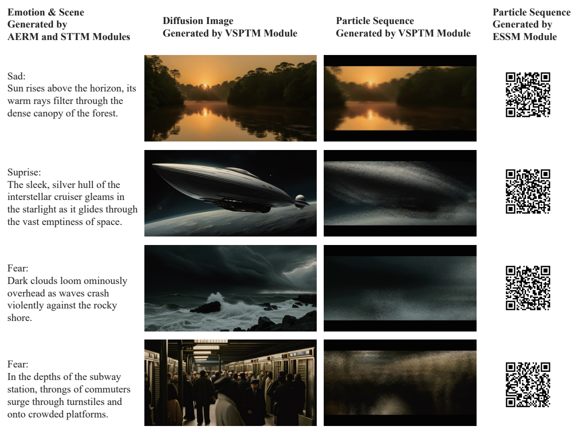

Figure 2: Visualization effect and Soundtrack of the Soundscape generated by EDAVS

- Audio Emotion Recognition Module (AERM) identifies emotional content in the user's voice. Raw input audio is preprocessed for analysis, features are extracted using signal processing and machine learning, then classified into emotion tags such as happiness, sadness, anger, etc. The fine-tuning model based on Wav2Vec2[Baevski et al. 2020] is used for Emotion Recognition[Enrique Hernández Calabrés 2024].
- Speech-to-Text Translation Module (STTM) converts input audio to text, informing image generation in VSPTM and thematic elements in ESSM. Whisper model[Radford et al. 2023] used for speech translation.
- Visual Synthesis and Particle Transformation Module (VSPTM) creates a visual representation of input audio content using text-to-image synthesis algorithms. Particle system influenced by emotional data from AERM, producing visual animations representing the user's emotional state. Fine-tuned Stable Diffusion model for better image quality.
- Emotive Soundscape Synthesis Module (ESSM) processes emotional tags and input audio text to generate auditory soundscape complementing visual output. AudioLDM 2[Liu et al. 2024] is used for high-fidelity soundscape audio generation.
- Integration and Output Module (IOM) merges outputs from VSPTM and ESSM, synchronizing audiovisual elements for cohesive user experiences, and aligning visual changes with emotive sound field. Real-time rendering provides interactive sessions where spoken emotions are seen and felt.

The visualization effect and soundscape soundtrack of EDAVS are shown in Fig. 2. The interaction environment of EDAVS comprises a high-quality microphone array, projector for dynamic visual particle animations, and surround sound speaker matrix as shown in Fig. 3. The immersive installation allows audience interaction with simulated scenarios. Fig. 4 shows our system's scene simulation. We constructed an art exhibition space, and the actual exhibition is presented in Fig. 5. Considerations include space layout, lighting, and ambient sound for enhancing experience and creating a seamless transition between real and virtual worlds.

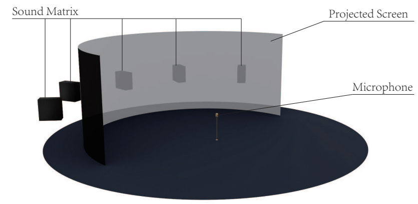

Figure 3: Hardware Distribution Diagram of the EDAVS System

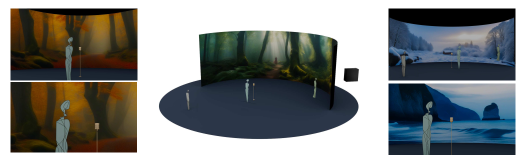

Figure 4: The Scene Simulation of the EDAVS System

Figure 5: Image Transformation Driven by the Soundscape

3 FUTURE WORKS

VR or AR technologies could be utilized to provide users with an even more immersive audio-visual experience. Additionally, as research in emotion visualization advances, we aim to further refine the dynamic effects representing emotions, enhancing the system's expressiveness and the user's sensory engagement.

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